[Pluedo-A Murder Mystery Game]

PROGRAMMING IN PYTHON-Project Submitted by

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Abstract

Pluedo is a Python-based mystery-solving application developed using the **Tkinter** library. The project challenges players to act as detectives across two distinct crime scenes to identify a culprit among multiple suspects. By analyzing victim data and cross-referencing suspect statements, the player must use deductive reasoning to solve the case. The implementation focuses on **event-driven programming** and **GUI-based state management**. While currently featuring two functional levels, this version serves as a foundational module for a more complex narrative-driven Python Project.

Problem Description

The project addresses the need for interactive educational tools that improve logical deduction. Standard text-based logic puzzles can be unengaging; **Pluedo** provides a visual interface where:

- **Level 1 & 2:** Each level presents a unique "Dead Person" scenario with a specific set of suspects.
- **Deduction Engine:** Players must compare conflicting suspect statements to identify logical inconsistencies.
- **Goal:** Successfully identify the culprit to progress or complete the game.

Libraries and Tools Used

Tkinter	Used to create the main game window, navigation buttons for Level 1 & 2, and the display area for suspect statements.
TinyDB	A lightweight, document-oriented database used to store and retrieve game data,such as suspect statements, victim profiles, and player progress, in a JSON format.

System Design / Workflow

The application follows a **Modular Scene-Switching** design. Instead of opening new windows for each level, the GUI dynamically updates the central container to reflect the current stage of the mystery.

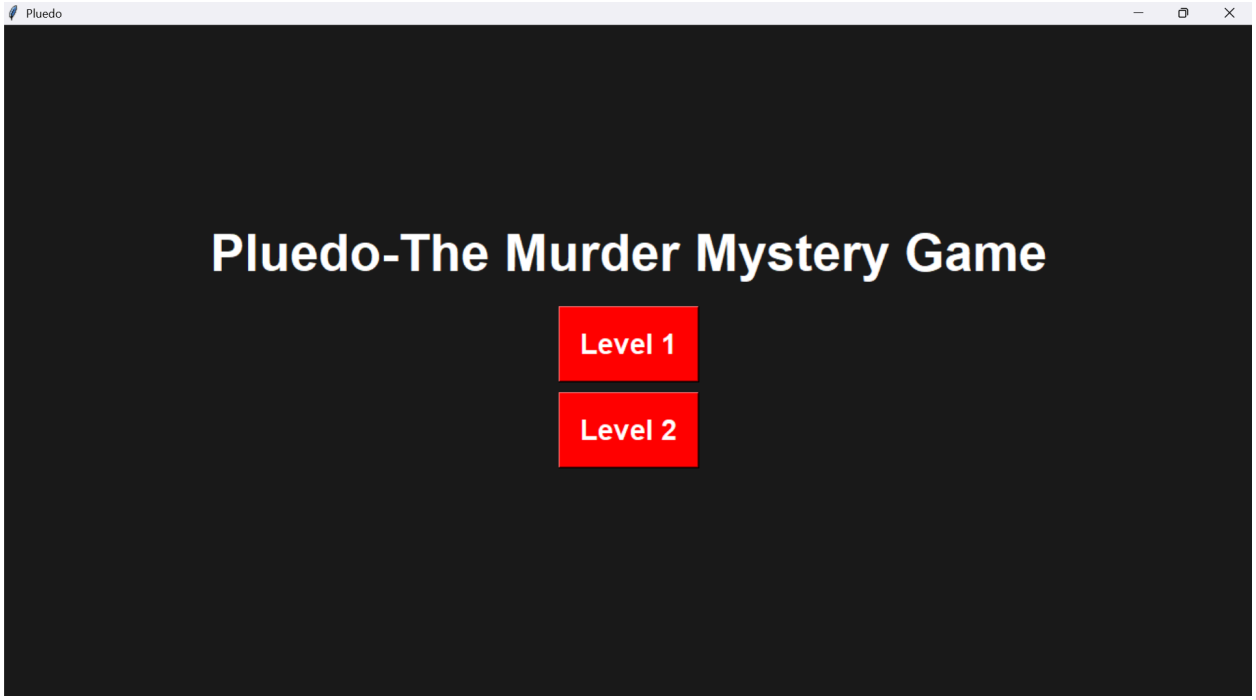
- **Initialization:** The game starts at the **Start Screen**, initializing variables for the player's progress.
- **Narrative Input:** For both levels, the system displays a "Crime Scene" overview (the dead person and the context) and a list of suspects.
- **The Interrogation Loop:** The user selects a suspect, triggering a function that fetches and displays that specific suspect's statement.
- **The Accusation Logic:** A validation check is performed when the user identifies a culprit. If correct, the workflow transitions to the next level's data set.

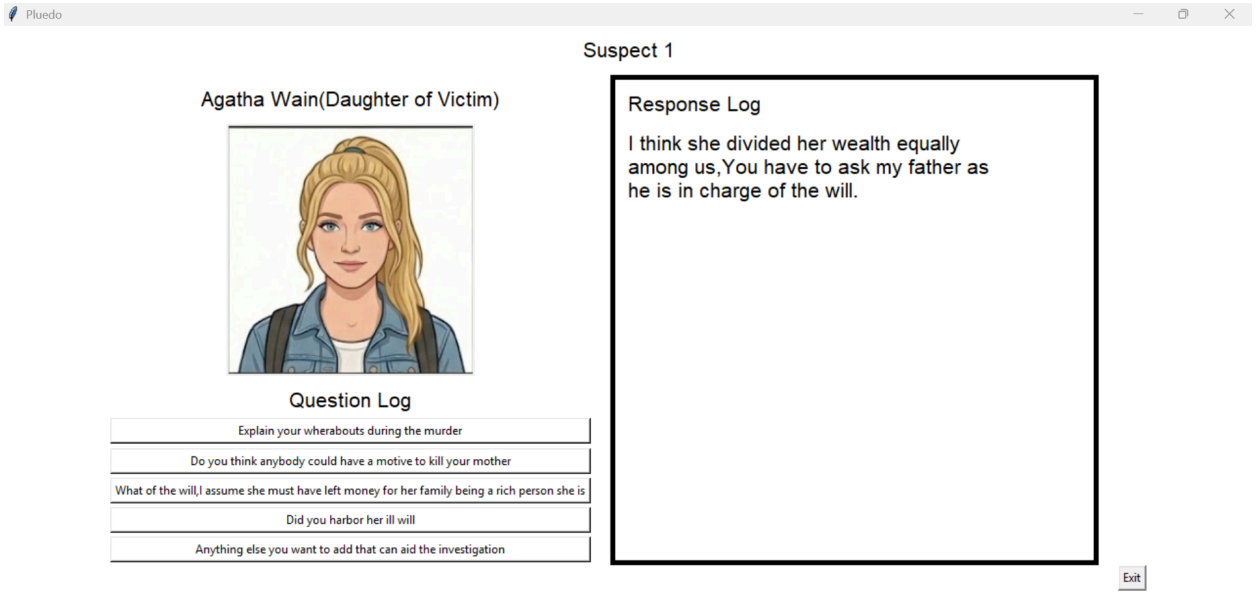
Implementation Details

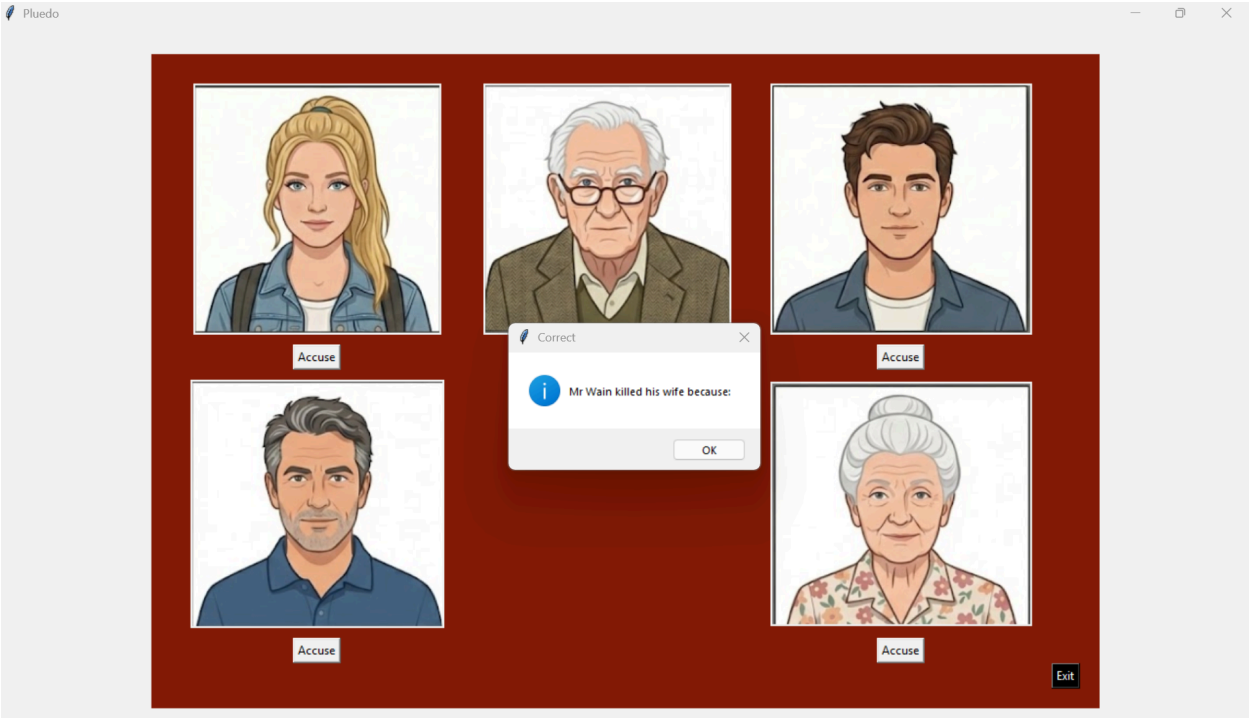
The project is built using a Functional Programming approach with a focus on UI stability.

- **GUI Framework (Tkinter)**: I used tkinter.Label for storytelling and tkinter.Button for player choices. The grid() manager was used to ensure the layout remains consistent across both levels.
- **Data Persistence with TinyDB**: To move beyond basic Python dictionaries and ensure data is handled professionally, **TinyDB** is used as the primary document-oriented database.
    - **Database Setup:** A JSON file (`db.json`) acts as the persistent storage for game data.
    - **Dynamic Content Loading:** Instead of hard-coding clues or suspect lists, the system queries TinyDB to fetch level-specific information.
    - **State Management:** TinyDB is used to track the player's progress, storing which clues have been found and which levels are unlocked.
- **State Management**: The game uses a "Level Tracker" variable. When Level 1 is solved, a function clears the existing widgets and repopulates them with the data (dead person, suspects, statements) for Level 2.
- **Error Handling**: Basic conditional logic ensures that players cannot bypass the interrogation phase without interacting with the suspects.

Results & Output


Pluedo
Pluedo-The Murder Mystery Game
Level 1
Level 2


Pluedo
Suspect 1
Agatha Wain(Daughter of Victim)
Question Log
Explain your wherabouts during the murder
Do you think anybody could have a motive to kill your mother
What of the will,I assume she must have left money for her family being a rich person she is
Did you harbor her ill will
Anything else you want to add that can aid the investigation
Response Log
I think she divided her wealth equally among us,You have to ask my father as he is in charge of the will.
Exit



Conclusion & Scope for Mini Project Extension

This individual project serves as a **foundational module** for a larger mystery engine.

Current Achievement:I have successfully implemented the core mechanics of a two-level deduction game using GUI elements and TinyDB for persistent data management".

Scope for Mini Project Extension (Future Work):

- **Refining Level 1:** I plan to improve the narrative flow of the first level by adding "clue icons" that provide hints before the player talks to suspects.
- **Complexity:** I will introduce a "Deduction Journal" to store gathered clues.
- **Persistence:** Adding a feature to save player progress between sessions using Python's file handling

References

Hasbro. (1949). *Cluedo* (Board Game). [Inspiration for game logic and "Whodunnit" narrative structure].
Python Software Foundation. *Tkinter — Python interface to Tcl/Tk*